

Fig. 1.12 The Gay-Berne potential depends strongly on the relative orientations of the rigid bodies, after a figure in [143]

in providing a precise parameterization [187]: using a general model for water (one not optimized for their specific experiment) the melting point of ice was underestimated by approximately 20 °K.

1.2 The N-Body Problem

In this book, we shall frequently use a compact, vectorial notation, where $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ represent vectors of the positions and velocities, and $\boldsymbol{M}$ is a diagonal mass matrix, so the equations of motion (1.2) become

$$\boldsymbol{M}\frac{d^2}{dt^2}\boldsymbol{q} = \boldsymbol{F}(\boldsymbol{q}) = -\nabla U(\boldsymbol{q}). \tag{1.3}$$

If the N particles are confined to the real line, then $\boldsymbol{q} = (q_1, q_2, \ldots, q_{\mathrm{N}})$ and the components of the force $\boldsymbol{F}$ are $F_i = -\partial U/\partial q_i$. For motion in $\mathbb{R}^3$, we write $\boldsymbol{q}_i = (q_{i,x}, q_{i,y}, q_{i,z})$ for the position vector of the ith particle, and the force acting on this particle is $\boldsymbol{F}_i = -\nabla_{\boldsymbol{q}_i} U = -\partial U/\partial \boldsymbol{q}_i$, that is, the vector with components $F_{i,x} = -\partial U/\partial q_{i,x}, F_{i,y} = -\partial U/\partial q_{i,y}, F_{i,z} = -\partial U/\partial q_{i,z}$. We may also write x_k, y_k, z_k for the x, y and z components of the position of particle k. We let $\mathrm{N_c}$ represent the total number of position coordinates of our system, thus for a system of N atoms in three space dimensions there are $\mathrm{N_c} = 3\mathrm{N}$ coordinates. In the typical case of 3-dimensional space, we view $\boldsymbol{q}$ as a $\mathrm{N_c} = 3\mathrm{N}$-dimensional vector and write $\boldsymbol{M} = \mathrm{diag}(m_1, m_1, m_1, m_2, m_2, m_2, \ldots, m_{\mathrm{N}}, m_{\mathrm{N}}, m_{\mathrm{N}})$.

The number of local directions in which the configurational (position) state can be varied is called the number of *degrees of freedom* $\mathrm{N_d}$. For the N-body system in $\mathbb{R}^3$ without additional constraints there are $\mathrm{N_d} = \mathrm{N_c} = 3\mathrm{N}$ degrees of freedom. If r independent constraints are present the number of degrees of freedom is $\mathrm{N_d} = \mathrm{N_c} - r$.

The total energy of the N-body system is a function of positions and velocities,

$$E(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}, \dot{\boldsymbol{q}}_1, \dot{\boldsymbol{q}}_2, \ldots, \dot{\boldsymbol{q}}_{\mathrm{N}}) = \sum_{j=1}^{\mathrm{N}} \frac{m_j \|\dot{\boldsymbol{q}}_j\|^2}{2} + U(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}). \tag{1.4}$$

Along the solutions of (1.3), the energy is conserved, since its derivative vanishes:

$$\frac{\mathrm{d}}{\mathrm{d}t}E = \sum_{j=1}^{\mathrm{N}} m_j \dot{\boldsymbol{q}}_j \cdot \ddot{\boldsymbol{q}}_j + \sum_{j=1}^{\mathrm{N}} \frac{\partial U}{\partial \boldsymbol{q}_j} \cdot \dot{\boldsymbol{q}}_j = \sum_{j=1}^{\mathrm{N}} \left(m_j \ddot{\boldsymbol{q}}_j + \frac{\partial U}{\partial \boldsymbol{q}_j} \right) \cdot \dot{\boldsymbol{q}}_j = 0.$$

As will become apparent in later chapters, it is sometimes necessary to introduce additional terms into the model to account for environmental conditions; these may destroy or alter the conservation of energy.

When we refer to the *value* of energy in a conservative system we typically write E instead of E (which represents the energy function).[3] Thus the positions and velocities of system evolving under Newtonian dynamics satisfy the equation

$$E(\boldsymbol{q}_1, \boldsymbol{q}_2, \ldots, \boldsymbol{q}_{\mathrm{N}}, \dot{\boldsymbol{q}}_1, \dot{\boldsymbol{q}}_2, \ldots, \dot{\boldsymbol{q}}_{\mathrm{N}}) = \mathrm{E}.$$

In addition to the energy, there may be other constants of motion which arise from the structure of the equations of motion. For example, if the full set of particles of an N-body system is included in the model, then the instantaneous force acting on particle i due to the presence of particle j will be the exact negative of the force acting on particle j due to particle i; this is a direct consequence of Newton's third law (every action has an equal and opposite reaction). This means that the sum of all the forces will vanish. Since

$$\frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \boldsymbol{F}_i,$$

we have

$$\sum_{i=1}^{\mathrm{N}} \frac{\mathrm{d}\boldsymbol{p}_i}{\mathrm{d}t} = \sum_{i=1}^{\mathrm{N}} \boldsymbol{F}_i = 0$$

and thus the three components of the *total momentum vector* $\boldsymbol{p}_{\mathrm{tot}} := \sum_{i=1}^{\mathrm{N}} \boldsymbol{p}_i$ will be conserved quantities. Additional first integrals sometimes arise in specific situations, depending on the form of the potential energy function.

For a few very special N-body systems, analytical solution is possible.

Example 1.3 (Harmonic Oscillator) Consider the system

$$\dot{x} = v, \qquad \dot{v} = -\Omega^2 x.$$

[3] We use roman (non-italic) letters for parameters that describe the thermodynamic state, e.g. number of degrees of freedom, the volume, the energy, the temperature or the pressure.